

User Manual: NCERT Physics Hybrid Graph-RAG Studio

1. Introduction

The NCERT Physics Hybrid Graph-RAG Studio is a sophisticated AI-powered tool designed to retrieve and synthesize information from NCERT Physics textbooks. It employs a hybrid approach combining dense vector search, sparse BM25 keyword matching, and a Knowledge Graph to provide grounded, highly relevant answers.

2. System Requirements

- Python 3.10 or higher
- At least 8GB RAM (16GB recommended for faster processing)
- Internet connection for initial setup (library installation & API access)
- A Google AI Studio API Key
- Windows/Linux/macOS compatible

3. Setup & Installation

3.1 Repository Setup

1. Clone the repository from your Git provider:

```
git clone <your-repo-url>
```

2. Navigate to the project directory:

```
cd hybrid_rag
```

3.2 Library Installation

Install the required Python dependencies using pip:

```
pip install -r requirements.txt
```

4. Data Preparation & Indexing

Before querying, you must ingest the PDF and build the hybrid indexes.

1. Place your 'NCERT-Class-12-Physics-Part-1.pdf' in the 'data/' directory.
2. Run the ingestion script to process the PDF into structured JSON:

```
python src/ingestion.py
```

3. Run the indexing script to build the Vector DB, BM25 Index, and Knowledge Graph:

```
python src/indexer.py
```

5. Starting the Application

1. Obtain a Google AI Studio API Key from <https://aistudio.google.com/>.

2. Start the application:

```
streamlit run src/app.py
```

3. Once the app launches in your browser, enter your API Key in the sidebar if prompted (or configure it via environment variables).

6. Querying the System

1. Enter your Physics-related question in the main text box.

2. The system will perform hybrid retrieval and show identified linked concepts from the Knowledge Graph.

3. View the grounded answer generated by the Gemini model, along with the source reference fragments.